Linear Algebra

Week 6

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1 Linear dependence and independence

Spanning answers the question, "Can these vectors produce every vector we need?" Linear independence asks a different question: "Have we included any redundant vectors?" A useful system of coordinates requires both properties.

Example 1.1 (Enough vectors, but one is redundant)**.** In $\mathbb{R}^3$, let

$$u = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \qquad v = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \qquad w = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}.$$

Since $w = u + 2v$, adding w does not enlarge $\operatorname{span}\{u, v\}$. Every vector in this span has the form $(a, b, a - b)^t$, so the three vectors do not span $\mathbb{R}^3$. If we add $e = (0, 0, 1)^t$, we can represent any $(a, b, c)^t$ as

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = au + bv + (c - a + b)e.$$

Thus (u, v, w, e) spans $\mathbb{R}^3$, but it still contains a redundant vector, namely w.

1.1 Definition and basic properties

Definition 1.1 (Linear dependence)**.** Let V be a vector space over $\mathbb{F}$, and let $v_1, \ldots, v_k \in V$. The list $(v_1, \ldots, v_k)$ is *linearly dependent* if there are scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$, not all zero, such that

$$\alpha_1 v_1 + \cdots + \alpha_k v_k = 0.$$

Otherwise the list is *linearly independent*: its only linear combination equal to zero has all coefficients zero.

The condition "not all zero" is essential. The all-zero coefficients always give the zero vector and say nothing about dependence.

We speak about lists because repetitions are allowed and, later, order will matter for coordinates. Reordering a list does not change its dependence or independence. A list with a repeated vector is dependent: subtract one occurrence from the other and assign coefficient zero to every remaining entry.

Example 1.2. The opening vectors satisfy $u + 2v - w = 0$, a nontrivial dependence relation. In contrast, the polynomials $1, x, \ldots, x^n$ are independent in $\mathbb{F}[x]$. Indeed,

$$\alpha_0 + \alpha_1 x + \cdots + \alpha_n x^n = 0$$

is equality of polynomials, so comparison of coefficients gives $\alpha_0 = \cdots = \alpha_n = 0$.

Theorem 1.2 (Basic tests for dependence)**.** For a finite list of vectors in a vector space:

1. if one entry is the zero vector, the list is dependent;
2. if a sublist is dependent, the whole list is dependent;
3. every sublist of an independent list is independent;
4. the list is dependent if and only if one vector is a linear combination of the others;
5. if $v_1 \neq 0$, the list $(v_1, \ldots, v_k)$ is dependent if and only if some v_j, with $j \geq 2$, is a linear combination of its predecessors $v_1, \ldots, v_{j-1}$;
6. a pair (u, v) is dependent if and only if one vector is a scalar multiple of the other.

Proof. For (1), use coefficient 1 on the zero vector and coefficient 0 elsewhere. For (2), extend a nontrivial relation by zero coefficients; (3) is the contrapositive.

For (4), a nontrivial relation has a coefficient $\alpha_j \neq 0$. Solving for that vector gives

$$v_j = -\alpha_j^{-1} \sum_{i \neq j} \alpha_i v_i.$$

Conversely, moving all terms of such a representation to one side gives a dependence relation with coefficient 1 on v_j.

For (5), take the largest index j with nonzero coefficient in a nontrivial relation. Then

$$v_j = -\alpha_j^{-1}(\alpha_1 v_1 + \cdots + \alpha_{j-1} v_{j-1}).$$

The case $j = 1$ would imply $v_1 = 0$, contrary to the hypothesis. The converse follows by moving terms to one side. Statement (6) is the two-vector case of (4), including the possibility that one vector is zero. □

Example 1.3 (Two directions)**.** The vectors $(1, 2, -1)^t$ and $(0, 1, 3)^t$ in $\mathbb{R}^3$ are independent. A multiple of the first vector with first coordinate zero must have multiplier zero, whereas the second vector is nonzero. Hence neither vector is a multiple of the other.

1.2 Testing independence with matrices

If A has columns $C_1, \ldots, C_n$, then

$$A\mathbf{x} = x_1 C_1 + \cdots + x_n C_n.$$

Thus the columns are dependent if and only if $A\mathbf{x} = 0$ has a nonzero solution. Equivalently,

$$\begin{aligned} \text{columns of } A \text{ independent} &\iff \operatorname{rank}(A) = n, \\ \text{columns of } A \text{ dependent} &\iff \operatorname{rank}(A) < n. \end{aligned}$$

Notice that the comparison is with the *number of columns*, not the number of rows.

Theorem 1.3 (Independence of rows under row reduction)**.** 1. Row-equivalent matrices have either both independent full row lists or both dependent full row lists.

2. The nonzero rows of an echelon matrix are independent.

Proof. An elementary row operation replaces the list of rows by an invertible linear recombination of that list. It therefore transforms a nontrivial relation into a nontrivial relation, in either direction. For example, under $R_i' = R_i + \alpha R_j$, $R_j' = R_j$, a relation with coefficients c_i, c_j becomes a relation in the old rows with coefficients $c_i, c_j + \alpha c_i$. This change of coefficients is invertible. Scaling by a nonzero scalar and interchanging rows are handled in the same way.

For (2), suppose a nontrivial combination of the nonzero echelon rows were zero. Choose the first row in that combination with nonzero coefficient. In its pivot column, later rows are zero and earlier coefficients are zero. The resulting entry is the nonzero coefficient times a nonzero pivot, a contradiction. □

Consequently, the full row list is dependent exactly when a row reduction produces at least one zero row. The same statement does not apply to the column list.

Example 1.4 (Zero rows do not automatically mean dependent columns)**.** The matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_1 - R_2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

has a dependent list of three rows, but its two columns are independent: its rank is 2, equal to the number of columns. The displayed replacement abbreviates the two successive elementary operations $R_3 \leftarrow R_3 - R_1$ and $R_3 \leftarrow R_3 - R_2$.

Example 1.5 (Finding a dependence relation)**.** Take

$$u = \begin{pmatrix} 1 \\ 2 \\ 0 \\ 3 \end{pmatrix}, \quad v = \begin{pmatrix} 0 \\ 1 \\ -1 \\ 2 \end{pmatrix}, \quad w = \begin{pmatrix} 2 \\ 7 \\ -3 \\ 12 \end{pmatrix}.$$

Put these vectors in the columns of a matrix:

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 2 & 1 & 7 \\ 0 & -1 & -3 \\ 3 & 2 & 12 \end{pmatrix}.$$

The operations $R_2 \leftarrow R_2 - 2R_1$ and $R_4 \leftarrow R_4 - 3R_1$ give

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & -1 & -3 \\ 0 & 2 & 6 \end{pmatrix}.$$

Now $R_3 \leftarrow R_3 + R_2$ and $R_4 \leftarrow R_4 - 2R_2$ give

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The rank is $2 < 3$, so the columns are dependent. The homogeneous equations are $x_1 + 2x_3 = 0$, $x_2 + 3x_3 = 0$. Choosing $x_3 = 1$ gives $-2u - 3v + w = 0$, or $w = 2u + 3v$.

A different valid approach puts the transposed given vectors in the rows of a matrix and tests row independence. Transposition preserves dependence relations between the listed vectors.

2 Bases and dimension

2.1 A basis combines spanning and independence

Definition 2.1 (A finite ordered basis)**.** A basis of V is an ordered finite list $B = (v_1, \ldots, v_n)$ that spans V and is linearly independent.

Unless stated otherwise, a basis means a finite ordered basis. The empty list is independent and its span is $\{0\}$. Thus the zero space has the empty basis, with no vectors. Under the more general convention used elsewhere in mathematics, bases may be infinite; that distinction will matter for $\mathbb{F}[x]$.

Example 2.1 (Standard coordinates in $\mathbb{R}^3$)**.** The standard vectors $e_1 = (1,0,0)^t$, $e_2 = (0,1,0)^t$, $e_3 = (0,0,1)^t$ form a basis, since every vector is uniquely

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = ae_1 + be_2 + ce_3.$$

For example, $(4,-2,7)^t = 4e_1 - 2e_2 + 7e_3$.

Theorem 2.2 (Standard bases)**.** Let $\mathbb{F}$ be a field.

1. The standard basis of $\mathbb{F}^n$ is $(e_1, \ldots, e_n)$, where $(e_i)_j = \delta_{ij}$.
2. The standard basis of $M_{m,n}(\mathbb{F})$ consists of the matrices E_{ij}, in row order, where
$$(E_{ij})_{kl} = \begin{cases} 1, & k = i \text{ and } l = j, \\ 0, & \text{otherwise.} \end{cases}$$
3. The standard basis of $\mathbb{F}_n[x]$, the polynomials of degree at most n together with the zero polynomial, is $(1, x, \ldots, x^n)$.

Proof. In each case every object is the sum of its entries or coefficients times the listed standard objects:

$$(a_1, \ldots, a_n)^t = \sum_{i=1}^{n} a_i e_i, \qquad A = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij} E_{ij}, \qquad p(x) = \sum_{j=0}^{n} a_j x^j.$$

These expansions establish spanning. A zero combination has every coordinate, matrix entry, or polynomial coefficient zero, establishing independence. □

Example 2.2 (A matrix space has many bases)**.** The usual basis of $M_2(\mathbb{C})$ is $(E_{11}, E_{12}, E_{21}, E_{22})$. Another basis is

$$(E_{11} + 2E_{12},\ E_{12},\ E_{21},\ E_{22}).$$

Indeed, any matrix can be represented as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a(E_{11} + 2E_{12}) + (b - 2a)E_{12} + cE_{21} + dE_{22}.$$

The formula also shows uniqueness by examining the four entries.

Example 2.3 (A nonstandard polynomial basis)**.** In $\mathbb{R}_2[x]$, consider

$$B = (1 + x,\ 1 - x,\ x + x^2).$$

To express $a + bx + cx^2$, solve

$$\alpha(1 + x) + \beta(1 - x) + \gamma(x + x^2) = a + bx + cx^2.$$

Comparing coefficients gives

$$\alpha + \beta = a, \qquad \alpha - \beta + \gamma = b, \qquad \gamma = c.$$

Hence

$$\alpha = \frac{a + b - c}{2}, \qquad \beta = \frac{a - b + c}{2}, \qquad \gamma = c.$$

The coefficients exist uniquely for every $a, b, c \in \mathbb{R}$, so this is a basis. For instance,

$$2 + 3x + x^2 = 2(1 + x) + 0(1 - x) + 1(x + x^2).$$

2.2 Why all bases have the same size

Theorem 2.3 (Comparison of sizes)**.** If S is a finite spanning list of a vector space V, and L is a finite independent list in V, then

$$|L| \leq |S|.$$

Proof. Write $S = (s_1, \ldots, s_m)$ and $L = (v_1, \ldots, v_k)$. If either list is empty, the claim is immediate: an empty spanning list implies $V = \{0\}$, which contains no nonempty independent list. Assume both are nonempty. Since v_1 is in the span of S and is nonzero, its representation uses some s_j with nonzero coefficient. Solving that representation for s_j shows that replacing s_j by v_1 leaves a spanning list of length m.

Repeat. Suppose $v_1, \ldots, v_r$ have already replaced r of the old spanning vectors. In the expression for v_{r+1} using the current spanning list, at least one coefficient on an unreplaced old vector must be nonzero. Otherwise v_{r+1} would lie in $\operatorname{span}\{v_1, \ldots, v_r\}$, contrary to independence. Solve for that old vector and replace it by v_{r+1}.

After m replacements, $(v_1, \ldots, v_m)$ spans V. If $k > m$, then v_{m+1} is a combination of those predecessors, again contradicting independence. Therefore $k \leq m$. □

Corollary 2.4 (Definition of dimension)**.** Any two finite bases of a vector space have the same number of vectors. This number is its dimension, denoted $\dim_{\mathbb{F}} V$ or simply $\dim V$ when the field is understood. In particular,

$$\dim\{0\} = 0, \qquad \dim \mathbb{F}^n = n, \qquad \dim M_{m,n}(\mathbb{F}) = mn, \qquad \dim \mathbb{F}_n[x] = n + 1.$$

Proof. Each basis is independent and the other basis spans. Apply Theorem 2.3 in both directions. □

Theorem 2.5 (The space of all polynomials has no finite basis)**.** The vector space $\mathbb{F}[x]$ cannot be spanned by finitely many polynomials.

Proof. If $p_1, \ldots, p_k$ are finitely many polynomials, choose a nonnegative integer d at least as large as the degree of each nonzero p_i; if all are zero, take $d = 0$. Every linear combination has degree at most d, unless it is zero. The polynomial x^{d+1} is therefore outside their span. □

Under the general definition allowing infinite bases, $\{1, x, x^2, \ldots\}$ is an infinite basis of $\mathbb{F}[x]$: each polynomial uses only finitely many monomials. Thus the precise conclusion here is the absence of a *finite* basis.

Example 2.4 (The field is part of the dimension)**.** The same matrix space can be viewed over different scalar fields:

$$\dim_{\mathbb{C}} M_2(\mathbb{C}) = 4, \qquad \dim_{\mathbb{R}} M_2(\mathbb{C}) = 8.$$

Over $\mathbb{C}$, use the four matrices E_{ij}. Over $\mathbb{R}$, use the eight matrices

$$E_{11}, E_{12}, E_{21}, E_{22},\ iE_{11}, iE_{12}, iE_{21}, iE_{22}.$$

Each complex entry has two real coefficients, its real and imaginary parts. The same distinction appears for a complex signal: $\mathbb{C}$ has basis (1) over $\mathbb{C}$, but basis $(1, i)$ over $\mathbb{R}$.

3 Coordinate vectors

3.1 Existence and uniqueness of coordinates

Theorem 3.1 (Unique representation in a basis)**.** Let $B = (v_1, \dots, v_n)$ be a basis of V. Every $v \in V$ has a unique representation

$$v = \alpha_1 v_1 + \cdots + \alpha_n v_n, \qquad \alpha_i \in \mathbb{F}.$$

Proof. Existence follows from spanning. If also $v = \beta_1 v_1 + \cdots + \beta_n v_n$, subtraction gives

$$(\alpha_1 - \beta_1)v_1 + \cdots + (\alpha_n - \beta_n)v_n = 0.$$

Independence implies $\alpha_i - \beta_i = 0$ for every i, proving uniqueness. □

Definition 3.2 (Coordinate vector)**.** The unique coefficients in this representation are the coordinates of v relative to B. Their column vector is

$$[v]_B = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \in \mathbb{F}^n.$$

The vector v may itself be a polynomial, a matrix, or another kind of object. Its coordinate vector is always a column with $n = \dim V$ scalar entries.

Example 3.1 (Coordinates of a polynomial)**.** For the ordered basis $B = (1 + x, 1 - x, x + x^2)$ in Example 2.3,

$$[2 + 3x + x^2]_B = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}.$$

Relative to the standard ordered basis $E = (1, x, x^2)$, the same polynomial has

$$[2 + 3x + x^2]_E = \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}.$$

The polynomial did not change; the chosen coordinate system did.

Example 3.2 (Coordinates of a matrix)**.** Use the ordered standard basis

$$E = (E_{11}, E_{12}, E_{13}, E_{21}, E_{22}, E_{23})$$

of $M_{2,3}(\mathbb{R})$. For

$$A = \begin{pmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{pmatrix},$$

we have

$$A = 2E_{11} - E_{12} + 4E_{13} + 0E_{21} + 3E_{22} + 5E_{23}, \qquad [A]_E = \begin{pmatrix} 2 \\ -1 \\ 4 \\ 0 \\ 3 \\ 5 \end{pmatrix}.$$

The order of the basis determines the order of these entries.

Example 3.3 (Do not omit zero coefficients)**.** In $\mathbb{R}_4[x]$, with $E = (1, x, x^2, x^3, x^4)$,

$$[2x^4 - 5x^2 + 3]_E = \begin{pmatrix} 3 \\ 0 \\ -5 \\ 0 \\ 2 \end{pmatrix}.$$

The zero entries retain the positions of the absent powers.

3.2 Coordinates preserve linear structure

Theorem 3.3 (Coordinate identities)**.** Let B be a basis of an n-dimensional space V, and let $w_1, \dots, w_k \in V$.

1. For all scalars $\alpha_1, \dots, \alpha_k$,
$$[\alpha_1 w_1 + \cdots + \alpha_k w_k]_B = \alpha_1 [w_1]_B + \cdots + \alpha_k [w_k]_B.$$
2. For $v \in V$,
$$v \in \operatorname{span}\{w_1, \dots, w_k\} \quad \Longleftrightarrow \quad [v]_B \in \operatorname{span}\{[w_1]_B, \dots, [w_k]_B\}.$$
3. The vectors $w_1, \dots, w_k$ are independent if and only if their coordinate vectors are independent.
4. The vectors $w_1, \dots, w_k$ span V if and only if their coordinate vectors span $\mathbb{F}^n$.

Proof. Write $B = (v_1, \dots, v_n)$ and $w_j = \sum_{i=1}^{n} c_{ij} v_i$. Then
$$\sum_{j=1}^{k} \alpha_j w_j = \sum_{j=1}^{k} \alpha_j \sum_{i=1}^{n} c_{ij} v_i = \sum_{i=1}^{n} \left(\sum_{j=1}^{k} \alpha_j c_{ij} \right) v_i.$$
Uniqueness of coordinates proves (1). Every column in $\mathbb{F}^n$ is the coordinate vector of exactly one vector of V, and $[v]_B = 0$ holds exactly when $v = 0$. Therefore (1) translates every span equation and every zero relation in both directions. This proves (2), (3), and (4). □

Example 3.4 (Testing polynomial independence by coordinates)**.** Consider in $\mathbb{R}_2[x]$
$$p_1 = 1 + x, \qquad p_2 = x + x^2, \qquad p_3 = 1 + 2x + x^2.$$
Their standard coordinate columns form
$$\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}.$$
The third column is the sum of the first two, so the polynomials are dependent. The relation is precisely $p_3 = p_1 + p_2$. This illustrates why matrix methods apply to vectors that are not originally columns of numbers.

4 Working with bases and dimension

4.1 Maximal independent lists and minimal spanning lists

Theorem 4.1 (Equivalent descriptions of a basis)**.** For a finite list $(v_1, \dots, v_k)$ in V, the following are equivalent:

1. the list is a basis;
2. it is independent and cannot be enlarged to a longer independent list in V;
3. it spans V and no proper sublist still spans V.

Proof. A basis is independent. Since it already spans V, any added vector is a combination of its entries and creates a dependence. Conversely, an independent list that does not span V can be enlarged by choosing w outside its span. In a relation $\sum_i \alpha_i v_i + \beta w = 0$, the case $\beta \neq 0$ would put w in that span; hence $\beta = 0$, and independence forces all other coefficients zero. Thus a maximal independent list spans.

A basis is also a minimal spanning list: if an entry could be removed without changing the span, that entry would be a combination of the others, contradicting independence. Conversely, a dependent spanning list contains a redundant entry by Theorem 1.2. Removing it leaves a smaller spanning list. Therefore a minimal spanning list is independent. □

Here maximal and minimal concern adding or removing vectors while retaining the stated property; they do not mean largest or smallest vector coordinates.

Example 4.1 (A parametrized subspace)**.** Find a basis and dimension of

$$W = \left\{ \begin{pmatrix} a \\ b \\ 3a-b \\ -a+2b \end{pmatrix} : a, b \in \mathbb{R} \right\}.$$

Separate the parameters:

$$\begin{pmatrix} a \\ b \\ 3a-b \\ -a+2b \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ 3 \\ -1 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ -1 \\ 2 \end{pmatrix}.$$

Thus W is the span of $u = (1,0,3,-1)^t$ and $v = (0,1,-1,2)^t$. They are independent: the first two coordinates of $\alpha u + \beta v = 0$ give $\alpha = \beta = 0$. Therefore (u, v) is a basis and $\dim W = 2$.

4.2 Six useful consequences of dimension

Theorem 4.2. Suppose $\dim V = n < \infty$.

1. Every list with more than n vectors is dependent.
2. No list with fewer than n vectors spans V.
3. An independent list of exactly n vectors is a basis.
4. A spanning list of exactly n vectors is a basis.
5. Every independent list in V can be extended to a basis.
6. Every finite spanning list contains a sublist that is a basis.

Proof. Statements (1) and (2) follow from the comparison theorem using a basis of length n.

For (5), start with an independent list. If it does not span, add a vector outside its span, retaining independence as proved above. This process stops by the time it has n vectors, since more than n independent vectors are impossible. When it stops, the list spans and is a basis. In particular an independent list already of length n cannot require another vector, proving (3).

For (6), repeatedly remove a redundant vector from a finite spanning list. The process terminates with a minimal spanning list, which is a basis. If the starting list has exactly n entries, removing any entry would leave fewer than n spanning vectors, impossible by (2). Hence it was independent already, proving (4). □

If a possibly infinite collection spans a finite-dimensional space, it also contains a finite basis: express a fixed finite basis using finitely many of its vectors, collect those vectors, and apply (6).

Example 4.2 (Extending one direction)**.** In $\mathbb{R}^2$, the nonzero vector $(2,1)^t$ is independent by itself. Add $(1,1)^t$. If

$$\alpha \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0,$$

then $2\alpha + \beta = 0$ and $\alpha + \beta = 0$. Subtracting gives $\alpha = 0$, then $\beta = 0$. The two vectors are independent and their number equals $\dim \mathbb{R}^2 = 2$, so they form a basis.

Example 4.3 (Removing a redundant sensor profile)**.** Suppose profiles in $\mathbb{R}^4$ are

$$u = \begin{pmatrix} 1 \\ 0 \\ 2 \\ -1 \end{pmatrix}, \quad v = \begin{pmatrix} 0 \\ 1 \\ -1 \\ 3 \end{pmatrix}, \quad w = \begin{pmatrix} 2 \\ 3 \\ 1 \\ 7 \end{pmatrix}.$$

Since $w = 2u + 3v$, it adds no direction to the span. The first two coordinates show u, v are independent. Thus (u, v) is a basis of the span of all three profiles, whose dimension is 2.

4.3 Dimension of a subspace

Theorem 4.3. If V is finite dimensional and $W \subseteq V$ is a subspace, then W is finite dimensional and

$$\dim W \leq \dim V.$$

Moreover, $\dim W = \dim V$ if and only if $W = V$.

Proof. Build an independent list inside W by repeatedly adding a vector of W outside its current span. Its length cannot exceed $\dim V$, so the process ends with a finite basis of W. This proves the inequality. If equality holds, that basis is an independent list of $\dim V$ vectors in V, so it spans V. Since it also spans W, the spaces are equal. The reverse implication is immediate. □

5 The dimension formula for a sum

Adding dimensions counts directions shared by the two spaces twice. The intersection measures exactly what must be subtracted.

5.1 A matrix-space example

Example 5.1 (A sum larger than either summand)**.** Inside $M_2(\mathbb{R})$, take

$$U = \left\{ \begin{pmatrix} a & b \\ 2a - b & a + 2b \end{pmatrix} : a, b \in \mathbb{R} \right\}, \qquad W = \{A \in M_2(\mathbb{R}) : A^t = A\}.$$

For U, separate parameters:

$$\begin{pmatrix} a & b \\ 2a - b & a + 2b \end{pmatrix} = a \underbrace{\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}}_{A_1} + b \underbrace{\begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}}_{A_2}.$$

These two matrices are independent by their top-row entries. Hence $\dim U = 2$.

Every symmetric matrix has the form

$$\begin{pmatrix} c & d \\ d & e \end{pmatrix} = c \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}_{B_1} + d \underbrace{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}_{B_2} + e \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}}_{B_3}.$$

Comparison of the entries shows independence, so $\dim W = 3$.

The combined five matrices span $U + W$. They cannot be independent in the four-dimensional space $M_2(\mathbb{R})$. To compute the dimension, put their standard coordinate vectors in the rows:

$$\begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & -2 & -1 \\ 0 & 0 & 2 & -2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Now $R_4 \leftarrow R_4 + R_3$, followed by $R_5 \leftarrow R_5 + \frac{1}{3}R_4$, gives

$$\begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & -2 & -1 \\ 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

There are four independent nonzero rows, so $\dim(U+W) = 4$ and $U + W = M_2(\mathbb{R})$. In fact (A_1, A_2, B_1, B_2) is a basis: the same reduction of those first four rows uses none of row 5 and leaves four nonzero rows. The omitted matrix is redundant because

$$A_1 + A_2 = B_1 + B_2 + 3B_3.$$

We can also find the shared direction directly. A matrix of U is symmetric exactly when

$$b = 2a - b \quad \Longleftrightarrow \quad a = b.$$

Therefore

$$U \cap W = \operatorname{span}\left\{ \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix} \right\}, \qquad \dim(U \cap W) = 1.$$

We have obtained $4 = 2 + 3 - 1$, illustrating the general formula.

5.2 Statement and proof

Theorem 5.1 (The first dimension formula)**.** Let U, W be finite-dimensional subspaces of a vector space V. Then

$$\boxed{\dim(U+W) = \dim U + \dim W - \dim(U \cap W).}$$

Proof. Choose a basis $(v_1, \ldots, v_r)$ of $U \cap W$. Extend it to a basis

$$(v_1, \ldots, v_r, u_1, \ldots, u_p) \quad \text{of } U$$

and to a basis

$$(v_1, \ldots, v_r, w_1, \ldots, w_q) \quad \text{of } W.$$

The combined list

$$(v_1, \ldots, v_r, u_1, \ldots, u_p, w_1, \ldots, w_q)$$

spans $U + W$, because it contains the vectors needed to expand any sum $u + w$.

To prove independence, suppose

$$\sum_{i=1}^{r} \alpha_i v_i + \sum_{j=1}^{p} \beta_j u_j + \sum_{k=1}^{q} \gamma_k w_k = 0.$$

Rearranging gives

$$\sum_{k=1}^{q} \gamma_k w_k = -\sum_{i=1}^{r} \alpha_i v_i - \sum_{j=1}^{p} \beta_j u_j \in U.$$

The left side also belongs to W, so it belongs to $U \cap W$. Hence it equals $\sum_i \delta_i v_i$ for suitable δ_i. The relation

$$\sum_{k=1}^{q} \gamma_k w_k - \sum_{i=1}^{r} \delta_i v_i = 0$$

is a relation in the displayed basis of W. Thus every $\gamma_k = 0$ and every $\delta_i = 0$. The original relation now lies in the basis of U, forcing every α_i and β_j to be zero. The combined list is independent and therefore a basis.

Its size is $r + p + q$. Since $\dim U = r + p$ and $\dim W = r + q$, this gives

$$\dim(U + W) = r + p + q = (r + p) + (r + q) - r.$$

□

Example 5.2 (Two planes sharing a line)**.** In $\mathbb{R}^3$, let

$$U = \operatorname{span}\{e_1, e_2\}, \qquad W = \operatorname{span}\{e_1, e_3\}.$$

Both spaces have dimension 2. Their intersection is $\operatorname{span}\{e_1\}$, of dimension 1, while $U + W = \mathbb{R}^3$. Thus $\dim(U + W) = 2 + 2 - 1 = 3$: the shared direction e_1 must be counted once.

Corollary 5.2 (When dimensions add)**.** Under the hypotheses of the dimension formula, $U + W$ is a direct sum if and only if

$$\dim(U + W) = \dim U + \dim W.$$

Proof. The sum is direct exactly when $U \cap W = \{0\}$, equivalently when $\dim(U \cap W) = 0$. Substitute in the formula. □

6 Row spaces, column spaces, and rank

6.1 Definitions and the meaning of consistency

Definition 6.1. For $A \in \mathbb{F}^{m \times n}$ with rows $R_1, \ldots, R_m$ and columns $C_1, \ldots, C_n$, define

$$\operatorname{Row}(A) = \operatorname{span}\{R_1, \ldots, R_m\} \subseteq M_{1,n}(\mathbb{F}),$$

$$\operatorname{Col}(A) = \operatorname{span}\{C_1, \ldots, C_n\} \subseteq \mathbb{F}^m.$$

The row space consists of row vectors with n entries. The column space consists of column vectors with m entries. They usually live in different ambient spaces, even though their dimensions will turn out to agree.

Example 6.1 (Two independent columns, three dependent rows)**.** Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 2 \end{pmatrix}.$$

Its first two rows are the standard basis of $M_{1,2}(\mathbb{R})$, so

$$\operatorname{Row}(A) = M_{1,2}(\mathbb{R}), \qquad \dim \operatorname{Row}(A) = 2.$$

Its columns are independent, and

$$\operatorname{Col}(A) = \left\{ \begin{pmatrix} a \\ b \\ a + 2b \end{pmatrix} : a, b \in \mathbb{R} \right\}, \qquad \dim \operatorname{Col}(A) = 2.$$

The third row is the first row plus twice the second. The three rows are therefore dependent, despite the independence of the two columns.

Corollary 6.2 (Column-space consistency test)**.** The system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b} \in \mathrm{Col}(A)$.

Proof. The equation is exactly $x_1C_1 + \cdots + x_nC_n = \mathbf{b}$, the definition of membership in the column span. □

6.2 Rows and columns of a matrix product

Theorem 6.3. Let $A \in \mathbb{F}^{m\times n}$ and $B \in \mathbb{F}^{n\times k}$. Then

$$\mathrm{Col}(AB) \subseteq \mathrm{Col}(A), \qquad \mathrm{Row}(AB) \subseteq \mathrm{Row}(B).$$

More precisely, every column of AB is a linear combination of the columns of A, and every row of AB is a linear combination of the rows of B.

Proof. Writing the columns of A as $C_1(A), \ldots, C_n(A)$, column j of the product is

$$C_j(AB) = \sum_{p=1}^{n} b_{pj} C_p(A),$$

because its ith entry is $\sum_{p=1}^{n} a_{ip}b_{pj} = (AB)_{ij}$. Similarly, if $R_p(B)$ are the rows of B, then

$$R_i(AB) = \sum_{p=1}^{n} a_{ip} R_p(B),$$

whose jth entry is the same defining sum for $(AB)_{ij}$. □

6.3 What row operations preserve

Theorem 6.4 (Row-space invariance)**.** If A and B are row equivalent, then

$$\mathrm{Row}(A) = \mathrm{Row}(B).$$

Proof. Every new row after an elementary operation is a combination of the old rows, giving one inclusion of row spaces. Reversibility gives the reverse inclusion. Repeat for the sequence of operations. □

Remark 6.5 (Column spaces can change)**.** Row operations do not generally preserve the column space. For example, over $\mathbb{R}$,

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} B = \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}.$$

Here

$$\mathrm{Col}(A) = \mathrm{span}\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}, \qquad \mathrm{Col}(B) = \mathrm{span}\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}.$$

Both have dimension 1, but they are different lines.

Theorem 6.6 (Dimension of the row space)**.** For every matrix A over a field,

$$\dim \mathrm{Row}(A) = \mathrm{rank}(A).$$

The nonzero rows of any echelon matrix row equivalent to A form a basis of $\mathrm{Row}(A)$.

Proof. Row reduction preserves the row space. In echelon form, the nonzero rows span that space and are independent by Theorem 1.3. Their number is the rank by definition. □

Example 6.2 (A basis for a span from echelon rows)**.** Let S be the span of

$$u = \begin{pmatrix} 1 \\ 0 \\ 2 \\ -1 \end{pmatrix}, \quad v = \begin{pmatrix} 0 \\ 1 \\ -1 \\ 3 \end{pmatrix}, \quad w = \begin{pmatrix} 2 \\ 3 \\ 1 \\ 7 \end{pmatrix}.$$

Put their transposes in rows and reduce:

$$\begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 3 \\ 2 & 3 & 1 & 7 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 3 \\ 0 & 3 & -3 & 9 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - 3R_2} \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The first two rows form a basis of the row space. Transposing them back gives the basis (u, v) of S, so $\dim S = 2$. The same method works for polynomial or matrix subspaces after replacing each object by its coordinate vector in a fixed basis.

6.4 Original pivot columns give a column-space basis

Theorem 6.7 (Choosing a column-space basis)**.** Row reduce A to an echelon matrix U. If its pivot columns have indices $j_1, \ldots, j_r$, then the columns of the *original matrix* A with these indices form a basis of $\mathrm{Col}(A)$. Consequently,

$$\dim \mathrm{Col}(A) = \mathrm{rank}(A).$$

Proof. The r pivot columns of U are independent: restricting to their pivot rows gives an upper triangular $r \times r$ matrix with nonzero diagonal, so back-substitution in a zero relation forces all coefficients zero. They span every column of U: use the pivots and back-substitution to match its entries in the r nonzero rows; all remaining rows are already zero.

The same sequence of row operations applies to every column. It is reversible, so a relation $\sum_j \alpha_j C_j(A) = 0$ holds if and only if $\sum_j \alpha_j C_j(U) = 0$. In particular, both independence and the expressions of nonpivot columns in terms of pivot columns transfer back to A. Thus the original pivot columns are independent and span the original column space. □

Example 6.3 (Keep the original columns)**.** Take

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 3 \end{pmatrix}.$$

Apply $R_3 \leftarrow R_3 - R_1$, then $R_3 \leftarrow R_3 - R_2$, and finally $R_1 \leftarrow R_1 - 2R_2$:

$$U = \begin{pmatrix} 1 & 0 & -2 & -3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot indices are $1, 2$. Therefore a basis of $\mathrm{Col}(A)$ is

$$\left(\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \right).$$

The corresponding columns of U, namely $(1,0,0)^t$ and $(0,1,0)^t$, form a basis of $\mathrm{Col}(U)$, a different space. The dependence coefficients do agree:

$$C_3(A) = -2C_1(A) + C_2(A), \qquad C_4(A) = -3C_1(A) + 2C_2(A).$$

For example, $-2(1,0,1)^t + (2,1,3)^t = (0,1,1)^t = C_3(A)$.

Theorem 6.8 (Row rank equals column rank)**.** For $A \in \mathbb{F}^{m\times n}$,

$$\dim \mathrm{Row}(A) = \dim \mathrm{Col}(A) = \mathrm{rank}(A) = \mathrm{rank}(A^t).$$

Proof. The two dimension equalities were proved above. Transposition identifies the columns of A with the transposes of the rows of A^t and preserves all relations between them. Hence

$$\dim \mathrm{Col}(A) = \dim \mathrm{Row}(A^t) = \mathrm{rank}(A^t).$$

□

Equal dimension does not make the row space and column space equal; their vectors have different shapes and generally different numbers of entries.

6.5 Rank inequalities for products

Corollary 6.9. For compatible matrices $A \in \mathbb{F}^{m\times n}$ and $B \in \mathbb{F}^{n\times k}$,

$$\mathrm{rank}(AB) \le \mathrm{rank}(A), \qquad \mathrm{rank}(AB) \le \mathrm{rank}(B).$$

If B is square and invertible, then

$$\mathrm{rank}(AB) = \mathrm{rank}(A).$$

Similarly, if P is an invertible $m \times m$ matrix, then

$$\mathrm{rank}(PA) = \mathrm{rank}(A).$$

Proof. The inclusions $\mathrm{Col}(AB) \subseteq \mathrm{Col}(A)$ and $\mathrm{Row}(AB) \subseteq \mathrm{Row}(B)$, together with equality of row rank and column rank, give both inequalities.

If B is invertible, apply the first inequality again to $A = (AB)B^{-1}$:

$$\mathrm{rank}(A) \le \mathrm{rank}(AB) \le \mathrm{rank}(A).$$

For an invertible P, use $A = P^{-1}(PA)$ in the same way, or recall that invertible left multiplication is a sequence of row operations. □